GGG-C-748b
AUGUST 21, 1959
SUPERSEDING
Fed. Spec. GGG-C-748a
May 5, 1954

FEDERAL SPECIFICATION

CUTTER, CIRCLE (ADJUSTABLE); FOR GASKETS AND WASHERS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers cutters (types I, II and III) for cutting circular washers and gaskets of materials such as asbestos, rubber, canvas, cork, leather, felt, and cardboard, in addition type III, style B will cut irregular shapes and strips; and cutters (type IV) for cutting holes or disks out of sheet metal, brass, aluminum, fiber, plastic, and wood.

1.1.1 *Federal specification coverage.* — Federal specifications do not include all the types and styles indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and styles.*—The cutters shall be of the following types and styles as specified (see 6.1) :

Type I.—Center pivoting.
  Style A.—Rectangular crossbeam.
  Style B.—Cylindrical crossbeam.
Type II.—End pivoting.
  Style A.—Adjustable pivot holder, machine adapter, and bit shank.
  Style B.—Nonadjustable pivot holder.
Type III.—Block, movable pivoting.
  Style A.—With extensions.
  Style B.—Without extensions.
Type IV.—Drill pivoting.
  Style A.—Bit shank.
  Style B.—Machine shank.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

OO-C-851 — Cutter Bit, Tool Holder (Solid High Speed Steel).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the

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Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.
MIL-P-15424—Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 — Marking for Shipment and Storage.
MIL-STD-130 — Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** — The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Governmental:

National Bureau of Standards Handbook:

H28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

Nongovernmental:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N. Y.)

3. REQUIREMENTS

3.1 **Materials.**—Materials used in the construction of circle cutters shall be as hereinafter specified for each type and style. All forgings or castings shall be of uniform quality and free from defects and imperfections that may affect their serviceability. When materials are used which are specifically designated, they shall be entirely suitable for the purpose intended.

3.1.1 *Illustrations.* — The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of cutters which are otherwise in accordance with this specification.

3.2 **Graduations of linear measurement scale.** — Graduation lines, when applicable, shall be of a uniform and readily distinguishable width, straight, permanent, and free from discontinuities and ragged edges. If the graduation lines are on metal or plastic members, they shall be sunk below the surface of the member and shall have sharp edges. Graduation lines representing equal fractional aliquots of any scale, in descending order, shall be of equal height and readily distinguishable by their relative height from graduation lines representing other fractional aliquots of the same scale. Lines representing inches shall be marked with the corresponding figures. The scale error for any linear setting on the members shall not exceed $\frac{1}{64}$ inch.

3.2.1 *Figures and markings.*—Figures and markings shall be clear, distinct, and permanent. Figures and markings on cutters having linear measurement scales shall be applied in a manner identical with that used to apply the graduation lines.

3.3 **Cutter blades.**—The cutter blades, except types III and IV (see 3.12.3 and 3.13.4), shall be of tool steel, hardened, and tempered from 40 to 45 on the Rockwell C scale (see 4.4.1). The cutter blades except types III and

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IV, shall have sharp round ends to facilitate a smooth and accurate cutting action. The blades and tool bits shall protrude at right angles (plus or minus 30 minutes) to the frame, beam, bar, or block, to insure correct setting and avoid chamfering the edges of the material being cut. Attachable cutter blades shall fit neatly in the holder with a minimum of play.

3.4 **Cutter-blade holders.**—The cutter-blade holders, except for types III and IV, shall be suitably constructed of steel or malleable iron and securely and readily clamped to the crossbeam by means of knurled thumbscrews, nuts, or setscrews. Means shall be provided for attaching the cutter blade securely to holder.

3.5 **Pivot.**—The pivot, except for types III and IV, (see 3.12.4 and 3.13.3) shall be of tool steel, suitably hardened and tempered, cylindrical in cross section and of proper length and size. The pivot shall be provided with a gradual taper at the point to assure penetration of the various thicknesses of material. The pivot shall protrude at right angles (plus or minus 30 minutes) to the beam or frame.

3.6 **Nuts, thumbscrews, and setscrews.**—The nuts, thumbscrews, and set screws shall be of steel, except for type III, which shall be provided with a threaded brass or steel insert, properly engaging the threads provided to lock the cutter blades or cutter blade holders to the beam, bar, or block. Screw threads shall be in accordance with Handbook H28.

3.7 **Repair parts.**—The successful bidders supplying cutters covered by this specification shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances, in order that such parts may be completely interchangeable, replaced, or adjusted without requiring modification.

3.8 **Marking.**—The cutters shall be marked in accordance with Military Standard MIL–STD–130, as practicable.

3.9 **Finish.**—Unless otherwise specified in the contract or order (see 6.1), the cutters shall be finished in accordance with the manufacturer's standard practice.

3.10 **Type I, center pivoting.**—Type I cutters shall consist of a crossbeam, bitstock pivoting body, pivot, cutter blades and cutter-blade holders. Type I cutters shall be capable of withstanding the test specified in 4.5.3.1.

3.10.1 *Crossbeam.*—The crossbeam shall be of steel or malleable iron, and graduated into increments of 8ths or 16ths of an inch. The numerals at the corresponding inch mark shall begin at zero at the center of the crossbeam with the numerals increasing toward the ends of the crossbeam.

3.10.2 *Bitstock pivoting body.*—The bitstock pivoting body shall be of steel or malleable iron, and integral with or securely attached to the crossbeam. The tapered square bit shank shall be of ¼ inch across the flats at the small end, 7/16 inch across the flats at the large end, by 1⅜ inches long. The round stock directly below the tapered shank shall be ⅜ inch in diameter by a minimum of ¾ inch long to facilitate a satisfactory fit and be held correctly in bitstock.

3.10.3 *Pivot.*—The pivot shall be firmly and rigidly secured by a setscrew, chuck, or forced fit, and shall be in accordance with 3.5. Pivots having a forced fit shall be capable of being removed for replacement or resharpening.

3.10.4 *Capacity.*—Type I cutters shall cut material up to and including ⅛ inch in thickness and from ¾- to, and including, 6-inch outside diameters and ⅝- to, and including, 5½-inch inside diameters.

3.10.5 *Style A, rectangular crossbeam.*—Style A cutters shall have a rectangular

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crossbeam with two integral cutting blades, or two cutting-blade holders with cutters. The crossbeam shall be slotted to provide a suitable passage and movement of the thumbscrews for adjusting the cutter blades or cutter-blade holders to the desired setting.

3.10.5.1 *Cutter blades.*—The cutter blades shall be integral with holder or attachable to cutter holder. Attachable cutter blades shall be in accordance with 3.3.

3.10.5.2 *Cutter-blade holders.*—The cutter-blade holders shall be in accordance with 3.4.

3.10.5.3 Type I, style A cutters shall be similar to figure 1.

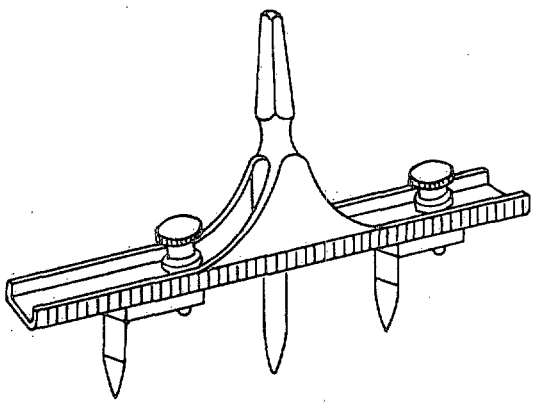


FIGURE 1.—Type I, style A, center pivoting, rectangular crossbeam.

3.10.6 *Style B, cylindrical crossbeam.*—Style B cutters shall have a cylindrical crossbeam, with two cutter-blade holders and two cutter blades. A flat surface shall be provided on the cross beam to accommodate the ends of the thumbscrews and to position the holders in proper alignment.

3.10.6.1 *Cutter blades.*—The cutter blades shall be securely attached to the cutter-blade holders and in alignment with the pivot. The cutter blades shall be in accordance with 3.3.

3.10.6.2 *Cutter-blade holders.* — The hole provided in the holder shall fit the beam without binding or excessive play and shall be in accordance with 3.4.

3.10.6.3 Type I, style B cutters shall be similar to figure 2.

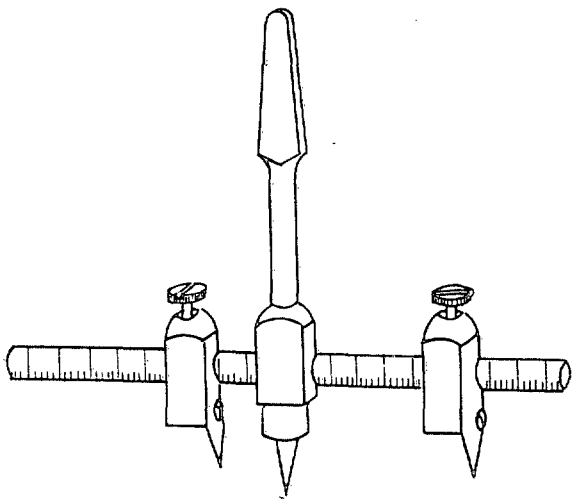


FIGURE 2.—Type I, style B, center pivoting, cylindrical crossbeam.

3.11 **Type II, end pivoting.**—Type II cutters shall be of steel and consist of a beam, pivot holder, pivot, cutter blade(s), and cutter-blade holder(s). The cutters shall be capable of withstanding the test specified in 4.6.3.1.

3.11.1 *Pivots.*—The pivots shall conform to 3.5 and be firmly and rigidly secured by a setscrew, chuck, or forced fit.

3.11.2 *Style A, adjustable pivot holder, machine adapter, and bit shank.*—Style A cutters shall have a square beam, two cutter blades, and pivot holder.

3.11.2.1 *Beam.*—One end of the beam shall be slotted to provide a suitable passage and movement of the cutter-blade holders to the desired setting.

3.11.2.2 *Pivot holder.*—The pivot holder shall be suitably constructed of a machine adapter shank, bit shank, and body. The machine adapter and bit shank shall be so con-

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structed as to assure a satisfactory fit and be held correctly in a drill press or a hand bit brace. Means shall be provided in the body for securely attaching the holder to the beam and the pivot to the holder. The hole provided in the holder shall fit the beam without binding or excessive end play.

3.11.2.3 *Cutter blades.*—The cutter blades shall be securely attached in the holders and in alignment with the pivot. The cutter blades shall be in accordance with 3.3.

3.11.2.4 *Cutter-blade holders.*—The cutter-blade holders shall be in accordance with 3.4.

3.11.2.5 *Capacity.* — The cutter shall cut materials up to and including ⅛ inch in thickness and from 1-inch up to, and including, 24-inch diameters.

3.11.2.6 Type II, style A cutters shall be similar to figure 3.

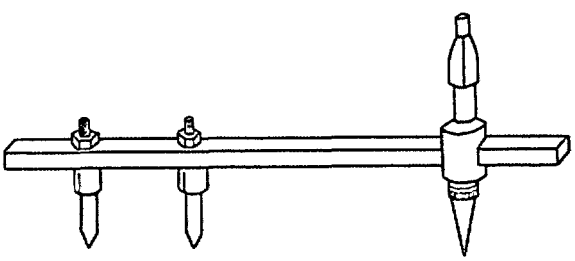


FIGURE 3.—Type II, style A, end pivoting, adjustable pivot holder, machine adapter and bit shank.

3.11.3 *Style B, nonadjustable pivot holder.* —Style B cutters shall have a rectangular beam, two cutter blades or cutter-blade holders, end pivot, and handles.

3.11.3.1 *Beam.*—The beam shall be graduated into increments of 8ths or 16ths of an inch. The numeral at the corresponding inch mark shall begin at the center of the end pivot with the numerals increasing toward the opposite end of the beam. An elongated slot of suitable width and length shall be provided for passage and movement of the thumbscrews and for adjusting and securing the cutter blades or cutter holders to the desired setting. Means shall be provided at one end of the beam, to permanently secure the end pivot.

3.11.3.2 *Handles.* — Handles shall be of clear, straight-grained hardwood such as maple, ash, hickory, or birch. Handles shall be round or suitably designed to facilitate a comfortable and secure handgrip. All exposed wooden surfaces shall be protected with a durable coating of enamel, varnish, shellac, or lacquer. Means shall be provided for attaching one handle permanently to the beam at the pivot end and the other handle by a thumbscrew to the outer cutter-blade holder.

3.11.3.3 *Cutter blades.*—The cutter blades shall be integral with holder or attachable to cutter-blade holder and in alignment with the pivot. Attachable cutter blades shall be in accordance with 3.3.

3.11.3.4 *Cutter-blade holders.*—The cutter-blade holders shall be in accordance with 3.4.

3.11.3.5 *Capacity.*—The cutter shall cut materials up to and including ⅛ inch in thickness and from 4-inches up to, and including, 20-inch diameters.

3.11.3.6 Type II, style B cutters, shall be similar to figure 4.

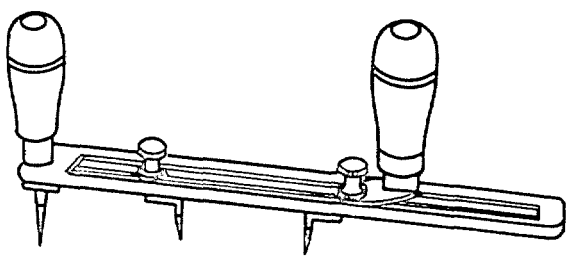


FIGURE 4.—Type II, style B, end pivoting, nonadjustable pivot holder.

3.12 **Type III, block, movable pivoting.**— Type II cutters shall consist of a block, pivot post with nut and washer(s), cutter blades,

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pivot pins, and cutting board with centering ferrule(s). A metal carrying case shall be provided to accommodate the cutter and its components, except the cutting board. Type III cutters shall be capable of withstanding the test specified in 4.6.3.2.

3.12.1 *Block.*—The block shall be either metal or phenolic with an elongated slot of suitable depth, width, and length to provide for housing, supporting and adjusting the pivot post to the desired setting. Suitable means shall be provided for locating and securing cutter blades in the end of the block. The bottom of the block, along the side of the pivot post slot, shall be graduated in $\frac{1}{32}$ inch increments, that, when set, will equal the actual diameter desired. The opposite side of the pivot post slot may be graduated in millimeters. The numerals at the marks shall start with zero at the cutter blade end of the pivot post slot, with the numerals increasing toward the opposite end. They shall otherwise conform to 3.2.

3.12.2 *Pivot post.* — The pivot post shall consist of a metal or phenolic body with a ferrule (properly inserted for centering the pivot pins), a stud screw permanently attached, fiber washer(s), and a suitable knurled nut for securing the pivot post to the block.

3.12.3 *Cutter blades.*—The cutter blades shall be properly formed to a sharp edge to facilitate a smooth and accurate cutting action. The blades shall be of high carbon steel, hardened and tempered to a hardness of from 90 to 93 on the Rockwell superficial 15N scale under a 15 kilogram (kg). load (see 4.5.1).

3.12.4 *Pivot pins.*—The pivot pins shall be of corrosion resisting steel or tool steel suitably hardened and tempered, cylindrical in cross section, of proper length and size. The pivot pin shall be provided with a gradual taper at the point to assure penetration of the various thicknesses of material, except for the nonpiercing pivot pin (see 3.12.7 and 3.12.8). The pivot pin shall protrude at right angles, plus or minus 30 minutes, to the base of the block.

3.12.5 *Cutting board.*—The cutting board shall have a plastic cutting surface, with a durometer hardness of 95 plus or minus 5 (see 4.6.2), with a suitable rigid backing for support and corrosion-resisting steel centering ferrule(s) inserted. The cutting surface shall be rough in order that gaskets will not slip while being cut. The plastic pad shall be removable and replaceable. The ferrule(s) shall be serrated to preclude turning or moving in the board and shall be perpendicular to the surface within plus or minus 30 minutes. Holes in ferrules shall fit pivot pins neatly or as required for a close running fit.

3.12.6 *Carrying case.*—A metal carrying case shall be provided to suitably accommodate the tool and components, except the cutting board. The case shall be of sheet metal of durable and rigid construction. The case shall be equipped with a hinged cover and a latching device. An instruction sheet containing clear and concise information as to the proper operation of the tool shall be included inside the carrying case. A parts list including manufacturer's replacement parts and identification numbers may be permanently attached to the top of the case on the inside.

3.12.7 *Style A, with extensions.*—Style A cutters shall be provided with three each of 0.017 and 0.025 inch thickness steel blades, two extension arms, three varied length pivot pins, one nonpiercing pivot pin for cutting solid discs or diaphragms, one pin handle for seating pin in material, one hexagon key wrench to tighten pivot post nut, one pivot post with two washers and a nut with a hexagon wrench hole, and a cutting board with two centering ferrules. The blade shall be held firmly by a plate and one screw with a knurled knob-type head.

3.12.7.1 *Capacity.* — The cutter shall be capable of cutting by successive cuts of at

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least ⅛ inch deep, when necessary, through pliable conventional gasket materials from 0.001 through 0.750 inch in thickness. Thick materials shall be cut through by cutting with the blade partially extended. Subsequent cuts are made after extending the blade farther until the blade cuts through 0.750 inch thick material. Style A cutters shall be capable of cutting diameters from 1 to 13 inches without extension arms and with extension arms added up to and including 36 inches in diameter. For diameters under 2 inches, the thickness of the material cannot exceed one-half the diameter in order to clear the blade.

3.12.7.2 *Body.*—The overall length of the body with the cutter head attached shall be not less than 7 inches. The width shall be not less than $1\frac{11}{16}$ inches.

3.12.7.3 *Cutter blades.*—The blades shall be provided with rounded points to facilitate a smooth and accurate cutting action. The blades shall otherwise conform to 3.12.3.

3.12.7.4 *Extension arms.*—Each extension arm shall be capable of adding an extra 12 inches to the diameter. They shall be interchangeable and capable of fitting onto the pivot post in the cutter body. They shall be capable of holding the pivot pin steady and true. Suitable means shall be provided to keep the extension arms in line and keep them from bending.

3.12.7.5 *Cutting board.*—The cutting board shall be at least 18 by 18 inches square and shall have two centering ferrules, one in the center for cutting diameters up to 18 inches and one ferrule in a corner for cutting larger diameters. The board shall otherwise conform to 3.12.5.

3.12.7.6 Type III, style A cutters shall be similar to figure 5.

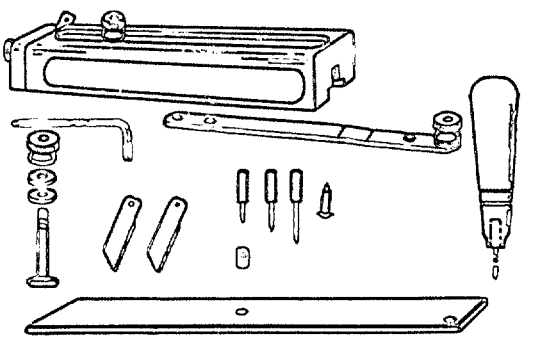


FIGURE 5.—Type III, style A, block, movable pivoting, with extensions.

3.12.8 *Style B, without extensions.*—Style B cutters shall be capable of following irregular lines and cutting circles, odd shapes, and strips. The cutter shall be furnished with six cutting blades (four pointed and two curved), six pivot pins (two short, two long, and two nonpiercing pivot pins for cutting discs), a pivot post with nut and washer, and a cutting board with one centering ferrule. The blade shall be held firmly by a plate and one screw with a knurled knob-type head.

3.12.8.1 *Capacity.* — The cutter shall be capable of cutting, by successive cuts of at least ⅛ inch deep when necessary, through pliable, conventional, gasket materials from 0.001 through 0.375 inch in thickness. Thick materials shall be cut through by cutting with the blade partially extended. Subsequent cuts are made after extending the blade farther until the blade cuts through 0.375 inch thick material. Style B cutters shall be capable of cutting diameters from ¼ inch to 6 inches. For diameters under 1 inch, the thickness of the material cannot exceed one-half the diameter in order to clear the blade.

3.12.8.2 *Body.*—The overall length of the body shall be not less than $4\frac{7}{16}$ inches. The width shall be not less than 1¼ inches. The body shall have a tracing index line aligned with the blade for following irregular lines.

3.12.8.3 *Cutting board.*—The cutting board shall be at least 6 by 6 inches square and shall have one centering ferrule. The board shall otherwise conform to 3.12.5.

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3.12.8.4 *Cutter blades.*—The cutter blades shall conform to 3.12.3 and be true within a tolerance of plus or minus 1/64 inch to the tracing index line on the block in order to follow irregular cutting lines.

3.12.8.5 Type III, style B cutters shall be similar to figure 6.

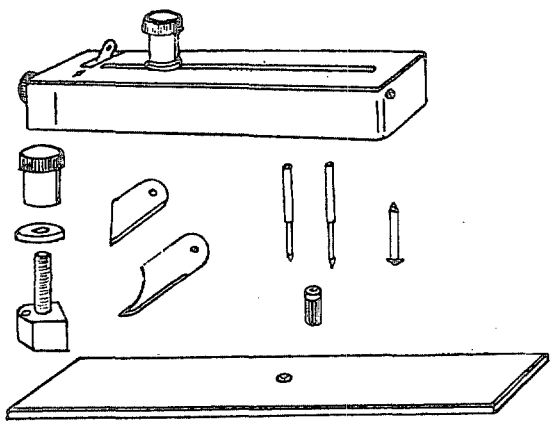


FIGURE 6.—Type III, style B, block, movable pivoting, without extensions.

3.13 Type IV, drill pivoting.—Type IV cutters shall consist essentially of an adjustable bar, body, drill, and tool bit.

3.13.1 *Adjustable bar.*—The adjustable bar shall be of good-quality steel properly heat-treated to resist wear or deformation. The bar shall be either circular, square, or rectangular in cross section. The circular bar shall be provided with a suitable flat surface of sufficient length to accommodate the end of the holding screw for securing and positioning the bar. Suitable means shall be provided at one end of the bar to properly accommodate and secure the tool bit parallel to the surface of the flat and at right angle to the longitudinal axis of the bar. The tool bit shall not become loose or detached under normal service conditions.

3.13.2 *Body.*—The body shall be of good-quality steel properly heat-treated to resist wear or deformation. The body shall be provided with a smooth hole or channel at right angle to the vertical axis for passage of the bar, and a smooth hole concentric with the axis of the body to accommodate the drill. Suitable means shall be provided in the body for locking the bar and drill. The bar and drill shall not become loose or detached under normal service conditions.

3.13.3 *Drill.*—The drill shall be of high-speed steel. Approximately 75 percent of the original flute length of the drill shall show a minimum hardness on the margin of 62 Rockwell C (150 kilogram load), or its equivalent (see 4.5.1). The minimum allowable hardness of the land at the shank end shall be 55 Rockwell C. The drill shall have a straight shank of a length suitable for the purpose intended.

3.13.4 *Tool bit.*—The tool bit shall be in accordance with type I, style 1, grade A, of Federal Specification OO–C–851. Type IV, style A cutters shall be provided with a 3/16-inch tool bit. Type IV, style B cutters shall be provided with a 1/4-inch tool bit.

3.13.5 *Style A, bit shank.*—Unless otherwise specified (see 6.1), style A cutters shall have a tapered squared bit shank suitable for use in a hand bit brace. The shank shall be 1/4 inch across the flats at the small end, 3/8 inch across the flats at the large end, by 1 1/4 inches long. Style A cutters shall be capable of withstanding the test specified in 4.6.3.3.

3.13.5.1 *Drill size.*—The drill size shall be 1/4 inch in diameter.

3.13.5.2 *Capacity.*—Style A cutters shall be capable of cutting mild sheet steel up to 16-gage (0.0598 inch) brass and aluminum, sheet fiber, plastic, or wood up to 3/8 inch in thickness. The cutters shall be capable of cutting from 1-inch up to, and including, 5 1/2-inch diameters.

3.13.5.3 Type IV, style A cutters shall be similar to figure 7.

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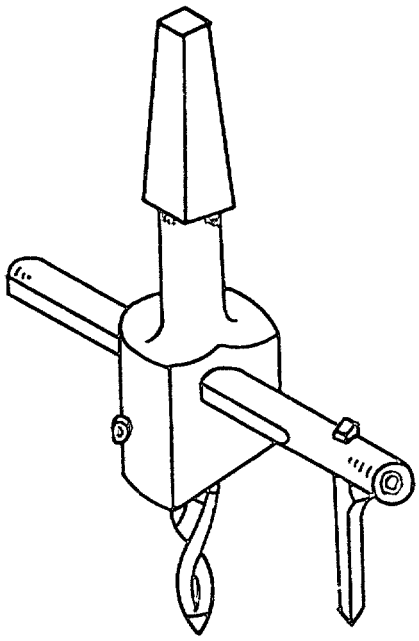


FIGURE 7.—Type IV, style A, drill pivoting, bit shank.

3.13.6 *Style B, machine shank.* — Unless otherwise specified (see 6.1), style B cutters shall have a ½-inch straight shank. Style B cutters shall be capable of withstanding the test specified in 4.6.3.4.

3.13.6.1 *Drill size.*—The drill size shall be either ¼ or ⅜ inch in diameter.

3.13.6.2 *Capacity.*—Style B cutters shall be capable of cutting sheet steel up to 10-gage (0.1345-inch), brass and aluminum, sheet fiber, plastic, and wood up to ¾ inch in thickness. The cutter shall be capable of cutting from 1⅞ inches up to and including 8 inches in diameter.

3.13.6.3 Type IV, style B shall be similar to figure 8.

3.14 Workmanship. — The workmanship shall be first class throughout. The cutters shall be free from rust, fins, burrs, external sharp or rough edges, corners, or surfaces, and defects which may impair serviceability or durability.

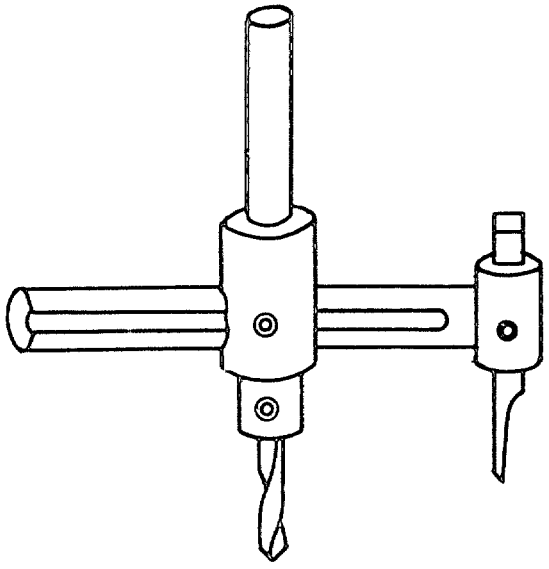


FIGURE 8.—Type IV, style B, drill pivoting, machine shank.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling for acceptance inspection.

4.2.1 *Inspection lot.*—All cutters of the same type and style offered for delivery at one time shall be considered a lot for purposes of acceptance inspection.

4.2.2 *Sampling for examination.*—A random sample of cutters shall be selected, in accordance with table I, from each inspection lot of material offered for Government examination of visual and dimensional characteristics with lot acceptance based on the following sampling requirements of table I and

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the procedures of Military Standard MIL–STD–105.

TABLE I.—*Sampling for visual and dimensional examination AQL (approx.) 1.5 percent defective*

Lot size (number of cutters)	Sample size (number to be examined) (reference Military Standard MIL–STD–105)
40 and under	Inspection level III
41 to 300	Inspection level II
301 and over	Inspection level I

4.2.3 *Sampling for tests.*—A random sample of cutters or cutting boards (when applicable) shall be selected from each inspection lot in accordance with the appendix to Military Standard MIL–STD–105 at inspection level L–8.

4.3 Visual and dimensional examination.—Each of the sample cutters selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification. Any cutter in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective cutters in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Classification of defects. — Inspection of the cutters shall be conducted as follows:

4.4.1 *Visual and dimensional examination.* —(Classification of Defects as per Military Standard MIL–STD–105).

Categories And Defects

AQL (approx.) 1.5 percent

100. *Critical:*
None defined.

200. *Major (sampling per table I).*

200.1 Type and style not as specified.

200.2 Incomplete, component parts missing.

200.3 Materials not as specified, not free of cracks, splits, or breaks.

200.4 Evidence of rust, fins, burrs, sharp or rough edges.

200.5 Graduations not uniform and readily distinguishable, not straight and permanent, or graduations not as specified. (except style A, type II and type IV).

200.6 Scale error exceeds the maximum allowable, (except type IV).

200.7 Bitstock pivoting body defective, not integral with (or securely attached to) the crossbeam (type I only).

200.8 Pivot defective, not of the proper length, not tapered, not firmly and rigidly secured, or not capable of being removed for replacement or resharpening.

200.9 Pivot holder defective, machine adapter and bit shank not as specified, holder not adjustable along the beam, holder binds or evidences excessive play, or cannot be locked in any position along the beam (type II only, except style B).

200.10 Pivot post defective, stud screw not securely attached or threads damaged, ferrule missing or improperly inserted, or nut fails to allow clearance for the stud screw when securing pivot post to block (type III only).

200.11 Pivot not of the proper length, or not as specified (type III only).

200.12 Cutter blades not sharp, not rigidly attached, or not as specified (type II only).

200.13 Cutter blades defective, not sharp, not rigidly attached, or not as specified (type I only).

200.14 Cutter blades defective, not double edged, not pointed, not sharp, or evidence of excessive play when cutter blades are attached to the box (type III only style B).

200.15 Cutter blade holders defective, cannot be readily clamped to the crossbeam, clamping screwhead (or nut) not knurled, or cutter blade holders stick or bind when moved along the beam (types I and II only).

200.16 Cutter blades defective, not formed as specified, not sharp, or evidence of excessive play when cutter blades are attached to the box (except style A, type III and type IV).

200.17 Handles not shaped as required, not coated, pivot handle not permanently attached, or other handle not attached by a thumbscrew (type II only except style A).

200.18 Capacity of cutter not as required.

200.19 Angle of cutter blades or pivot with beam not within the specified tolerance (except type IV).

200.20 Body defective, cannot be locked to the bar,

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body cannot be moved along the bar, or evidence of sticking, binding, or excessive looseness, hole or channel not right angle to vertical axis (type IV only).

200.21 Drill not sharp, not the required length and size, or drill not as specified (type IV only).

200.22 Tool bit not sharp, not the required size, not as specified, or tool bit not securely attached to bar (type IV only).

200.23 Size of machine shank not as specified (type IV only, except style A).

300. *Minor (sampling per table I)*:

300.1 Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanent.

300.2 Tapered square bit shank dimensions not as required (types I and IV only).

300.3 Length of round stock (below tapered shank) not as specified (type I only).

4.5 Tests.—Each of the sample cutters or cutting boards selected in accordance with 4.2.3 shall be subjected to the hardness and performance tests specified in 4.6.1 through 4.6.3.4, as applicable to verify compliance with this specification. Any sample cutter or cutting board which does not conform to these characteristics shall cause rejection of the lot represented by the sample.

4.5.1 *Acceptance tests (possible tests failures)*.

(a) Hardness:

(1) Hardness of cutter blades not within the specified values on the Rockwell C scale (types I, II and III only).

(2) Hardness of drill less than the specified minimum values on the Rockwell C scale (tye IV only).

(b) Performance test:

(1) Failure of cutter to cut the required number of washers, circles not concentric, not the same size, or cuts not clean and positive when cutting the specified material, cutter breaks, deforms, chips or cracks (types I, II and III only).

(2) Failure of cutter to cut the required number of disks, disks not concentric, not the same size, or cuts not clean and positive when cutting the specified material, cutter breaks, deforms, chips or cracks (types IV only).

4.6 Test procedures.

4.6.1 *Hardness test.*—Each sample cutter selected from each inspection lot in accordance with 4.2.3 and the cutter blades shall be subjected to a hardness test to determine conformance with 3.3, 3.12.3 and 3.13.3.

4.6.2 *Durometer hardness test for cutting board.*—Each sample cutting board selected from each inspection lot in accordance with 4.2.3, shall be subjected to a durometer hardness test to determine conformance with 3.12.5.

4.6.3 *Performance test.*—Each sample cutter selected from each inspection lot in accordance with 4.2.3 shall be subjected to the applicable test specified in 4.6.3.1, 4.6.3.2, 4.6.3.3 and 4.6.3.4.

4.6.3.1 *Types I and II.*—Each sample cutter shall be tested by cutting 10 washers of the same diamter from stiff cowhide leather of ⅛ inch thickness. The sample cutters shall consistently cut concentric circles identical in size within plus or minus $\frac{1}{32}$ inch, and show a clean and positive cut (see 3.3).

4.6.3.2 *Type III.*—Each sample cutter shall be tested by cutting 10 washers of the same diameter from ⅛ inch thick sheet asbestos. The blade shall be extended sufficiently to cut entirely through the test specimen on the first cut. The sample cutters shall consistently cut concentric circles identical in size within plus or minus $\frac{1}{32}$ inch, and show a clean and positive cut (see 3.3 and 3.12).

4.6.3.3 *Type IV, style A.* — Each sample cutter shall be tested by cutting 10 disks of

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the same diameter out of 16-gage (0.0598-inch) mild sheet steel. The sample cutters shall consistently cut concentric disks identical in size within a plus or minus 1/64 inch, and show a clean and positive cut (see 3.13.5).

4.6.3.4 *Type IV, style B.* — Each sample cutter shall be tested by cutting 10 disks of the same diameter out of 10-gage (0.1345 inch) sheet steel. Sample cutters shall consistently cut concentric disks identical in size within plus or minus 0.005 inch, and show a clean and positive cut (see 3.13.6).

4.7 Rejected lots.—Rejected lots may be offered again for inspection provided the contractor has repaired or removed all nonconforming cutters. The inspector shall again select and examine samples from such resubmitted lots to verify compliance with this specification.

4.8 Inspection of preparation of delivery.—Sample cutters and packs shall be selected and inspected in accordance with Military Specification MIL–P–116 to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

(Civil agencies, Federal Standard 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment).

5.1 Cleaning, preservation, and packaging.

5.1.1 *Level A.*—Cutters shall be cleaned, preserved, and packaged in accordance with level A of Military Specification MIL–P–15424.

5.1.2 *Level C.*—Cutters shall be cleaned, preserved and packaged in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Cutters packaged as specified (see 6.1) shall be packed in accordance with the level A requirements of Military Specification MIL–P–15424.

5.2.2 *Level B.*—Cutters packaged as specified (see 6.1) shall be packed in accordance with the level B requirements of Military Specification MIL–P–15424.

5.2.3 *Level C.*—Cutters packaged as specified (see 6.1) shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Container or method of shipment shall be in accordance with the carrier rules or regulations as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military marking.* — In addition to any special marking required by the contract or order or herein, interior and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

5.3.2 *Non-Military marking.*—Marking for shipment shall be in accordance with the standard marking of Federal Standard No. 123.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type and style required (see 1.2).

(*c*) Whether a particular finish is required (see 3.9).

(*d*) Whether shank other than that specified is required (see 3.13.5 and 3.13.6).

(*e*) Selection of applicable level of preservation and packaging and the level of packing required (see 5.1 and 5.2).

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6.1.1 When repair parts for cutters are required, the following should be specified:

(*a*) Manufacturer's name or trademark of cutter for which required.

(*b*) Name of part.

(*c*) Quantity of each item.

(*d*) Any other data applicable to cutter which will assist in securing the proper repair parts.

6.2 Transportation description.—Transportation description applicable to these itms is:

Tools, not otherwise indexed by name.
Carload weight 30,000 pounds.

Hand tools, not otherwise indexed.
Truckload volume weight 30,000 pounds, subject to Rule 34, National Motor Freight Classification.

6.3 Illustrations.—The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of cutters which are otherwise in accordance with this specification.

6.4 Items procured under this specification for Military use are to be limited to the variety shown on the applicable Military Standards. Personnel of the Military Departments are requested to refer to these documents for guidance.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:
Army—Ordnance Corps
Navy—Bureau of Ships
Air Force.

Copies of this specification may be purchased for 10 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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RICHMOND, VA. 23212

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